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Smart Campus Announcement Assistant

RESEARCH QUESTION

Does fine-tuning BERTurk classify campus announcements more accurately (Macro-F1) than a TF-IDF + logistic-regression baseline, on announcements written outside our generation templates?

WHAT A "NO" LOOKS LIKE

BERTurk gains **< 0.03 Macro-F1**, or the paired-bootstrap 95% CI of the gain includes 0 → a linear model is enough for stage 01, and the project's effort should go to extraction and QA.

CHANGED SINCE THE PROPOSAL — AND WHY

Old: "How can NLP techniques turn scattered university announcements into structured, searchable information and answer students' questions?"
Why: an open "how" question has no "no" result, and on our template-based test set TF-IDF already scores **1.000**. We kept the system and narrowed the question to the one comparison that is still open: out-of-template text.

DATA AS LOADED

420 labelled announcements + 24 hand-written, test set frozen

Set	Source	N	TR / EN
Train	Synthetic “campus world” scripts/generate_dataset.py, seed 42	**252**	198 / 54
Dev		**84**	66 / 18
Test A (frozen)		**84**	**71 / 13**
Test B (frozen)	**Hand-written, real-style, never trained on**	**24**	**19 / 5**
GTÜ raw (unlabelled)	gtu.edu.tr, 24 departments, collected Nov 11 scripts/collect_gtu.py	**818**	≈ all TR

6 classes: exam 92 · event 76 · registration 71 · cancellation 63 · room change 59 · scholarship 59. Split: stratified 60/20/20, seed 2026, ids in `data/splits/midproject_split.json` (sha256 `2cafd21d7400`). Preprocessing: 420 → clean_text 420 → dedup 420 (0 removed). 1,292 entity spans (DATE/TIME/LOCATION/COURSE).

Licence: all text is team-written (generator + hand-written set) → released CC BY 4.0; GTÜ announcements are public pages © GTÜ: we release only URLs + our labels, not the text.

ONE REAL EXAMPLE (ANN-0001, TRAIN)

Title: KIM107 dersi iptal
Text: “1 Kasım 2026 tarihinde yapılması planlanan KIM107 Genel Kimya dersi, öğretim üyesinin rahatsızlığı nedeniyle yapılmayacaktır.”

category = cancellation
entities = DATE “1 Kasım 2026” → 2026-11-01 · COURSE “KIM107” → KIM 107
summary = “KIM107 dersi 1 Kasım tarihinde yapılmayacak.”

HONEST CAVEAT

Train/dev/test A are synthetic, so they share vocabulary with our rules and templates. That is why Test B exists and why a real test set (Test C) is the biggest item on the plan.

BASELINE RESULT

Test A is saturated; out-of-template text is where the gap is

System (trained on train, 252)	Dev Macro-F1	Test A Macro-F1 [95% CI]	Test A Acc	Test B Macro-F1 [95% CI]	Test B Acc
Majority class	0.059	0.059 [0.04, 0.08]	0.214	0.048 [0.01, 0.09]	0.167
Keyword rules	0.851	0.840 [0.76, 0.91]	0.857	0.801 [0.60, 0.94]	0.792
TF-IDF + NB	1.000	1.000 [1.00, 1.00]	1.000	0.789 [0.56, 0.94]	0.792
TF-IDF + linear SVM	1.000	1.000 [1.00, 1.00]	1.000	0.874 [0.68, 1.00]	0.875
TF-IDF + LogReg (main baseline)	**1.000**	**1.000** [1.00, 1.00]	**1.000**	**0.874** [0.68, 1.00]	**0.875**

0.874

TF-IDF + LogReg on hand-written Test B (n = 24)

0.126

drop from Test A to Test B

Spread: all baselines are deterministic, so the interval is a **95% bootstrap CI** (1,000 resamples of the test set, seed 0). Test B's CI is ~0.3 wide: with 24 items we cannot resolve a 0.03 difference → we need a larger real test set.
Reproduce: `python evaluation/midproject.py`

ERROR ANALYSIS

65 baseline errors in 7 categories

#	Category	N	Example (gold → predicted)
E1	Topic beats action exam room change → exam	**21**	*"27 Kasım 2026 tarihinde yapılacak CENG 213 mazeret sınavı Kütüphane Seminer Salonu yerine Kültür Merkezi salonunda ya…"* **room_change → exam**
E4	No keyword fires falls back to a default class	**16**	*"The double major applications period starts on November 18, 2026 and ends on November 22, 2026. All steps are complet…"* **registration → exam**
E2	Shared cue "başvuru" registration → scholarship	**13**	*"Akademik takvim gereği çift anadal başvuruları 18.11.2026 - 22.11.2026 tarihleri arasında yapılacaktır. Süresi içinde…"* **registration → scholarship**
E3	Registration about an exam "exam applications" → exam	**7**	*"The English proficiency exam applications period starts on November 1, 2026 and ends on 5 November 2026. All steps ar…"* **registration → exam**
E5	Event mentions exams "sınav haftası öncesi …" → exam	**3**	*"Sınav haftası öncesi moral ve motivasyon için Doğa Yürüyüşü düzenliyoruz. Tarih: 17/11/2026, Saat: 15:00, Yer: Amfi 1…"* **event → exam**
E6	Room name = event cue "Seminer Salonu" → event	**2**	*"Kapasite nedeniyle CENG 301 dersi 17.11.2026 itibarıyla Kütüphane Seminer Salonu dersliğinden D Blok 105 dersliğine t…"* **room_change → event**
E7	TF-IDF: unseen wording hand-written set only	**3**	*"Join our guest lecture on large language models for Turkish on Dec 3 at 4 pm in Conference Hall. Everyone is welcome."* **event → cancellation**

Keyword rules: 45 train + 12 dev + 5 Test B errors (rules are not trained, so train errors are fair); TF-IDF: 3 Test B errors. Test A errors were not inspected (frozen). File: `reports/midproject_errors.csv`

ERROR ANALYSIS

What the errors tell us

1 · THE LABEL IS THE ACTION, NOT THE TOPIC (E1, E3, E5 = 31)

“MAT 102 sınavı Konferans Salonu yerine E-205'te” is a room change, but every cue word says exam. Bag-of-words counts cannot weigh “X yerine Y” against “sınav”. A contextual encoder can — this is the strongest argument for BERTurk.

2 · VOCABULARY GAPS (E4, E7 = 19)

“Don't miss…”, “no lecture”, “ekle-sil”, “guest lecture”, “tuition waiver”, “ilişik kesme”: the hand-written set uses words that never appear in training. TF-IDF fails here; pretrained subword knowledge may not.

3 · AMBIGUOUS SHARED CUES (E2, E6 = 15)

“başvuru” is both scholarship and registration; “Seminer Salonu” is a room, not a seminar. TF-IDF learns these from data (0 errors on Test A) — so these are a rules problem, not a learning problem.

4 · EXTRACTION (STAGE 02) ON TEST B: 7 ERRORS

Date ranges without a year (“15-17 Mayıs” → wrong year, 4), free-form places (“Fen Fakültesi amfileri”, 2), dotted time “17.30'da” (1). Micro-F1 is still 0.947.

SO

Expect BERTurk to help on categories 1 and 2 only, which exist mainly outside the templates → measure on Test B and a real Test C, not on Test A.

METHOD STATUS

BERTurk fine-tuning implemented and run on CPU

THE ONE CONTROLLED CHANGE

Replace **TF-IDF features + LogReg** with a **fine-tuned BERTurk encoder** (`dbmdz/bert-base-turkish-cased`). Same split, same input (title + text), same labels, same metric.

AS IMPLEMENTED

4 epochs · lr 3e-5 · batch 16 · max length 128 · weight decay 0.01 · seeds 13, 42, 2026 · `python evaluation/midproject.py --with-berturk`
Compute: 5.8 min per seed on a laptop CPU; final runs (BERTurk + XLM-R, 3 seeds, larger train) ≈ 2 GPU-hours on Colab T4.

Dev set (n = 84)	Macro-F1	Accuracy
Keyword rules	0.851	0.857
TF-IDF + LogReg	1.000	1.000
BERTurk (mean ± std, 3 seeds)	**0.807 ± 0.134**	**0.841 ± 0.100**

First finding: BERTurk is **below** TF-IDF on dev and unstable across seeds (per seed: 0.918, 0.618, 0.885). Training loss is still 1.1–1.6 after 64 steps → under-trained, not a ceiling effect. Next: 10–20 epochs, lr 5e-5, epochs chosen on dev, on GPU. Test sets untouched until Dec 9.

REST OF THE PIPELINE (ALREADY WORKING)

Extraction rules: micro-F1 0.947 on Test B · Summaries: structured template ROUGE-L 0.953 vs lead-1 0.432 · QA (50 questions, end-to-end with predicted labels): 50/50 answers correct · Flask web app + CLI.

PLAN TO THE FINAL

Remaining experiments, owners, dates

Step	Owner	Date
~~Collect ≥ 300 real announcements~~ **done: 818 from 24 GTÜ departments**	Helin S.	Nov 11 ✓
Double-annotate 150 GTÜ items (+ "other" class) → **Test C**, κ on 50, freeze	Helin S. + Nisanur K.	Nov 27
BERTurk, 3 seeds on GPU, choose epochs on dev	Cevahir A.	Nov 24
Ablation: XLM-R vs BERTurk; title removed	Cevahir A.	Dec 4
Fix extraction errors X1–X3; hybrid NER on real data	Nisanur K.	Dec 4
QA on 50 real student questions	Emre K.	Dec 7
One test run of all systems on A/B/C + paired bootstrap	Cevahir A.	Dec 9
Error analysis of the method	Nisanur K.	Dec 11
Final report draft (paper format) + demo	Emre K. (all)	Dec 18

PROTOCOL (FIXED TODAY, BEFORE METHOD RESULTS)

Macro-F1 (primary), accuracy · frozen tests A, B (+C on Nov 27) · 3 seeds for neural models · 95% bootstrap CI · **a gain is real only if ΔMacro-F1 ≥ 0.03 and the paired-bootstrap CI excludes 0**.

BIGGEST RISK → FALLBACK

Real GTÜ text is far from our labels: median classifier confidence 0.89 on Test A vs **0.37 on the 818 GTÜ items**, and many are none of our 6 classes. Labelling may be slow / κ low.
Fallback: label 150 instead of 300, add an "other" class, report the minimum detectable difference.

PROPOSAL FEEDBACK

RQ narrowed to a testable yes/no; baselines + error analysis added before any neural model.

APPENDIX A · PROJECT CARD 1/4

Fields 1–3

1 · TITLE AND ABSTRACT

Working title: Do We Need BERTurk to Sort Campus Announcements? Linear Baselines vs. Fine-tuned Encoders on Out-of-Template Turkish Notices

Draft abstract (116 words): University announcements arrive through websites, e-mail and club channels, and students miss exam, room and deadline changes. We build a four-stage assistant (classification, information extraction, summarisation, cited question answering) for Turkish and English announcements and ask whether a fine-tuned Turkish encoder is needed for the first stage. On 420 labelled synthetic announcements a TF-IDF + logistic-regression baseline reaches 1.000 Macro-F1, but only 0.874 on 24 hand-written announcements; most baseline errors confuse the topic of an announcement with the action it asks for. Fine-tuned BERTurk reaches **[GAP: Test B / Test C Macro-F1 ± CI]** , which **[GAP: does / does not]** exceed our 0.03 threshold. The extraction rules reach 0.947 micro-F1 and end-to-end QA answers 50/50 questions.

2 · RESEARCH QUESTION

RQ: Does fine-tuning BERTurk classify campus announcements more accurately (Macro-F1) than TF-IDF + logistic regression on announcements written outside our generation templates?

"No" result: ΔMacro-F1 < 0.03 or the paired-bootstrap 95% CI includes 0 → the linear model suffices.

Changed: proposal asked "How can NLP techniques turn scattered announcements into structured, searchable information and answer questions?" — no falsifiable outcome, and the in-template test is saturated (TF-IDF 1.000). The system is unchanged; the question is narrowed to stage 01 on out-of-template text.

3 · CONTRIBUTIONS (TESTABLE CLAIMS)

- **C1.** On out-of-template announcements (Test B/C), fine-tuned BERTurk beats TF-IDF + LogReg by ≥ 0.03 Macro-F1 (paired bootstrap). Refuted if the CI includes 0.
- **C2.** The gain comes from "topic-vs-action" errors (E1/E3/E5): BERTurk removes ≥ 50% of them while E2/E6 errors are already 0 for TF-IDF.
- **C3.** A Turkish-only encoder (BERTurk) is at least as good as multilingual XLM-R on the 20% English announcements (ablation). Refuted if XLM-R is ≥ 0.03 better on English items.

Fields 4–6

4 · RELATED WORK

- Devlin, J., Chang, M.-W., Lee, K., Toutanova, K. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. *NAACL-HLT*. — We fine-tune a BERT-style encoder on a small 6-class domain task, not on general benchmarks.
- Schweter, S. (2020). BERTurk – BERT models for Turkish. Zenodo, doi:10.5281/zenodo.3770924. — Our method model; we test whether its gains hold on short, mixed TR/EN campus notices.
- Conneau, A., Khandelwal, K., Goyal, N., et al. (2020). Unsupervised Cross-lingual Representation Learning at Scale. *ACL*. — XLM-R is our ablation for the English share of announcements.
- Wang, S., Manning, C. D. (2012). Baselines and Bigrams: Simple, Good Sentiment and Topic Classification. *ACL*. — Motivates a strong linear baseline; we add character n-grams for Turkish suffixes.
- Joulin, A., Grave, E., Bojanowski, P., Mikolov, T. (2017). Bag of Tricks for Efficient Text Classification. *EACL*. — Same claim that linear models are competitive; we test it on low-resource Turkish text.
- Lewis, P., Perez, E., Piktus, A., et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *NeurIPS*. — Our QA retrieves and filters structured fields and cites sources instead of training a generator.
- Dror, R., Baumer, G., Shlomov, S., Reichart, R. (2018). The Hitchhiker's Guide to Testing Statistical Significance in NLP. *ACL*. — Basis for our paired-bootstrap decision rule.
- Dodge, J., Ilharco, G., Schwartz, R., et al. (2020). Fine-Tuning Pretrained Language Models: Weight Initializations, Data Orders, and Early Stopping. arXiv:2002.06305. — Why we report 3 seeds for every neural model.

5 · DATA

Source: synthetic "campus world" generator (`scripts/generate_dataset.py` , seed 42, 4 source types) + 24 hand-written real-style announcements (team-authored, CC BY 4.0) + **818 raw GTÜ announcements** from 24 departments (`scripts/collect_gtu.py` , public pages © GTÜ → we share URLs + labels only).

Counts: generated 420 → clean_text 420 → dedup 420 → split train 252 / dev 84 / test A 84; Test B 24. 335 TR / 85 EN; 1,292 entity spans. GTÜ: 818 collected (517 pages, 259 PDF titles, 42 links; 0 duplicates, 3 timeouts) → 150 to be labelled as Test C.

Example: "1 Kasım 2026 tarihinde yapılması planlanan KIM107 Genel Kimya dersi … yapılmayacaktır." → cancellation; DATE 2026-11-01, COURSE KIM 107.

Guideline: `docs/annotation_guidelines.md` (label = the student's required action). **Agreement:** not measured yet — κ on a 50-item double-annotated sample of the real set, due Nov 27.

6 · METHOD AND BASELINE

Baseline as run: TF-IDF (word 1–2-grams + char 3–5-grams, folded Turkish) + logistic regression (C = 5); also keyword rules, NB, SVM, majority.

Method as implemented: BERTurk (`dbmdz/bert-base-turkish-cased`) fine-tuned, 4 epochs, lr 3e-5, batch 16, max length 128; dev Macro-F1 0.807 ± 0.134 (3 seeds) vs 1.000 for TF-IDF — under-trained after 64 steps, so epochs will be tuned on dev.

The one change: the text representation + classifier (sparse TF-IDF + LogReg → fine-tuned encoder); data, input fields, split and metric are identical.

Compute: baselines < 10 s CPU; BERTurk 5.8 min/seed on CPU. Needed: ≈ 2 GPU-hours (Colab T4) for BERTurk + XLM-R × 3 seeds on the larger train set.

Fields 7–9

7 · EVALUATION PROTOCOL (FIXED NOV 11, BEFORE METHOD RESULTS)

Metric: Macro-F1 over 6 classes (primary); accuracy (secondary). Extraction: micro P/R/F1 on (label, normalised value).

Frozen tests: A = 84 synthetic (ids + sha256 2cafd21d7400 in `data/splits/`); B = 24 hand-written; C = 150 real, frozen on Nov 27 before any model sees it. Each system is run on the tests once (Dec 9).

Spread: 3 seeds (13, 42, 2026) for neural models, mean ± std; 95% bootstrap CI (1,000 resamples, seed 0) for all.

Smallest real difference: ΔMacro-F1 ≥ 0.03 *and* paired-bootstrap 95% CI of Δ excludes 0.

8 · RESULTS SO FAR (FINAL RESULTS TABLE)

System	Test A Macro-F1 [CI]	Test B Macro-F1 [CI]	Test C
Majority class	0.059 [0.04, 0.08]	0.048 [0.01, 0.09]	*Dec 9*
Keyword rules	0.840 [0.76, 0.91]	0.801 [0.60, 0.94]	*Dec 9*
TF-IDF + NB	1.000 [1.00, 1.00]	0.789 [0.56, 0.94]	*Dec 9*
TF-IDF + linear SVM	1.000 [1.00, 1.00]	0.874 [0.68, 1.00]	*Dec 9*
TF-IDF + LogReg (main)	**1.000 [1.00, 1.00]**	**0.874 [0.68, 1.00]**	*Dec 9*
BERTurk (3 seeds)	*Dec 9*	*Dec 9*	*Dec 9*
XLM-R base (ablation)	*Dec 9*	*Dec 9*	*Dec 9*
BERTurk, no title (ablation)	*Dec 9*	*Dec 9*	*Dec 9*

Baselines are deterministic, so the CI comes from bootstrap; seed std applies to neural rows.

9 · ERROR ANALYSIS — 65 BASELINE CLASSIFICATION ERRORS (+ 7 EXTRACTION ERRORS)

#	Category	N	Example (gold → predicted)
E1	Topic beats action	**21**	*"27 Kasım 2026 tarihinde yapılacak CENG 213 mazeret sınavı Kütüphane Seminer Salonu yerine Kü…"* **room_change → exam**
E4	No keyword fires	**16**	*"The double major applications period starts on November 18, 2026 and ends on November 22, 20…"* **registration → exam**
E2	Shared cue "başvuru"	**13**	*"Akademik takvim gereği çift anadal başvuruları 18.11.2026 - 22.11.2026 tarihleri arasında ya…"* **registration → scholarship**
E3	Registration about an exam	**7**	*"The English proficiency exam applications period starts on November 1, 2026 and ends on 5 No…"* **registration → exam**
E5	Event mentions exams	**3**	*"Sınav haftası öncesi moral ve motivasyon için Doğa Yürüyüşü düzenliyoruz. Tarih: 17/11/2026,…"* **event → exam**
E6	Room name = event cue	**2**	*"Kapasite nedeniyle CENG 301 dersi 17.11.2026 itibarıyla Kütüphane Seminer Salonu dersliğinde…"* **room_change → event**
E7	TF-IDF: unseen wording	**3**	*"Join our guest lecture on large language models for Turkish on Dec 3 at 4 pm in Conference H…"* **event → cancellation**
X1–3	Extraction: year-less date range / free-form place / "17.30'da"	**4/2/1**	*"konserler 15-17 Mayıs tarihlerinde Kampüs Açık Hava Tiyatrosu'nda"* → 2026 instead of 2027, place missed

APPENDIX D · PROJECT CARD 4/4

Fields 10–12

10 · RESEARCH LEAD SECTION — NISANUR KARA (RESEARCH LEAD)

Uludoğan, G., Yirmibeşoğlu Balal, Z., Akkurt, F., Türker, M., Güngör, O., Üsküdarlı, S. (2024). TURNA: A Turkish Encoder-Decoder Language Model for Enhanced Understanding and Generation. ***Findings of ACL 2024*****.** — Evaluates Turkish models on general benchmarks and finds encoder models such as BERTurk strong on classification; we test whether that holds for a tiny, noisy, mixed-language domain where a linear model is already near its ceiling.

Edwards, A., Camacho-Collados, J. (2024). Language Models for Text Classification: Is In-Context Learning Enough? ***LREC-COLING 2024*****.** — Shows fine-tuned smaller models often beat prompted LLMs on classification (English); we ask the prior question, whether fine-tuning beats a linear model at all, in Turkish, and can add a prompted Claude row as an extra comparison.

Ablation we will run: (a) XLM-R base vs BERTurk with the identical recipe (published multilingual baseline); (b) BERTurk without the title field; (c) learning curve with 25 / 50 / 100% of train. All on frozen tests A/B/C, 3 seeds.

11 · PLAN AND RISK

Real data: done Nov 11 (818 GTÜ items) · Test C double annotation + κ, freeze — Helin S. + Nisanur K. Nov 27 · BERTurk GPU runs — Cevahir A. Nov 24 · XLM-R / no-title ablation — Cevahir A. Dec 4 · extraction fixes + hybrid NER — Nisanur K. Dec 4 · QA on 50 real questions — Emre K. Dec 7 · single test run + paired bootstrap — Cevahir A. Dec 9 · method error analysis — Nisanur K. Dec 11 · report draft — Emre K. (all) Dec 18.

Biggest risk: real GTÜ text is far from our label set (median confidence 0.37 vs 0.89 on Test A; many items fit none of the 6 classes), so labelling is slow or κ is low. **Fallback:** 150 labelled items, an added "other" class, and the minimum detectable difference reported as a limitation.

Proposal feedback: RQ made falsifiable with an explicit "no"; baselines and error analysis added first.

12 · REPRODUCIBILITY

Repository: https://github.com/Cevahir11/smart-campus-assistant

One command (field 8): `python evaluation/midproject.py` (add `--with-berturk` for the method dev row; `python evaluation/error_analysis.py` for field 9).

Seeds: data 42 · split 2026 · bootstrap 0 · neural 13 / 42 / 2026. **Versions:** Python 3.12.6, scikit-learn 1.9.1, numpy 2.5.4, torch 2.15.0 (CPU), transformers 5.22.2.

Who did what: Helin S. — data collection, guideline; Nisanur K. — research lead (related work, ablation design), extraction, error analysis; Cevahir A. — classifiers, evaluation protocol, repository; Emre K. — summarisation, QA, web demo.

AI tools: Claude (Anthropic) via Claude Code helped write evaluation scripts and draft slide text; every number was produced by running the scripts above and checked by the team.